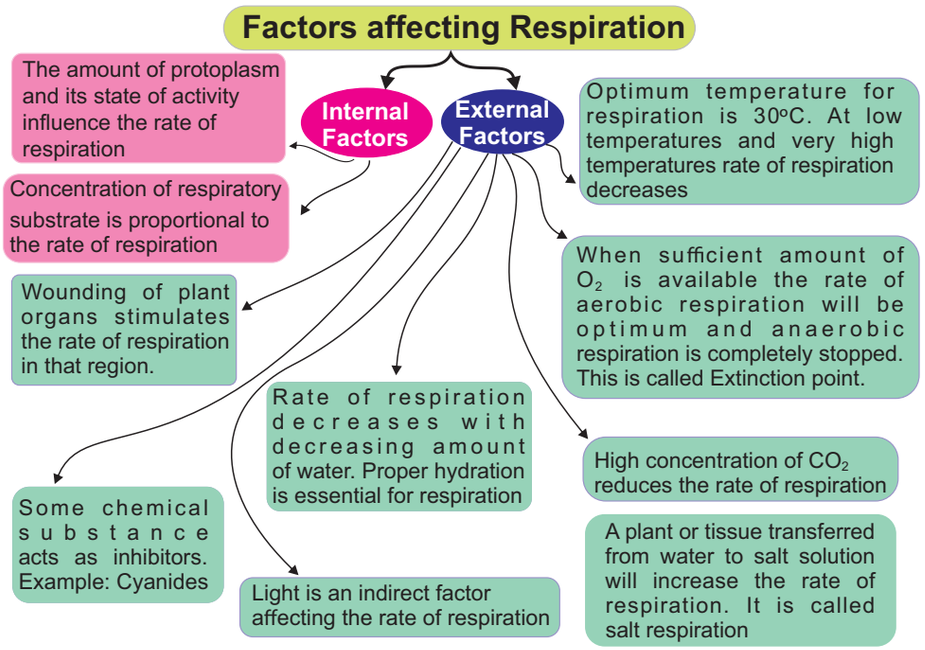
## 14.8 Factors Affecting Respiration




**Check your grasp!**

- Why Microorganisms respire anaerobically?
- Does anaerobic respiration take place in higher plants?

## 14.9 Pentose Phosphate Pathway (Phospho Gluconate Pathway)

During respiration breakdown of glucose in cytosol occurs both by glycolysis (about 2/3) as well as by oxidative pentose phosphate pathway (about 1/3). Pentose phosphate pathway was described by **Warburg**, **Dickens** and **Lipmann** (1938). Hence, it is also called **Warburg-Dickens-Lipmann pathway**. It takes place in cytoplasm of mature plant cells. It is an alternate way for breakdown of glucose (Figure 14.15).

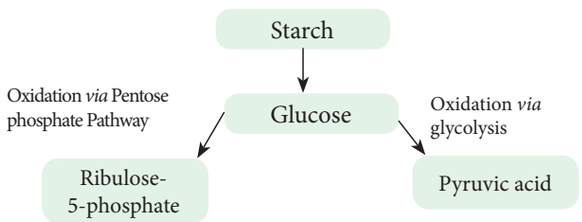



**Figure 14.15:** Fate of Glucose in HMP shunt and Glycolysis

It is also known as **Hexosemonophosphate shunt (HMP Shunt)** or **Direct Oxidative Pathway**. It consists of two phases, oxidative phase and non-oxidative phase. The oxidative events convert six molecules of six carbon Glucose-6-phosphate to 6 molecules of five carbon sugar Ribulose-5 phosphate with loss of $6CO_2$ molecules and generation of 12 NADPH + $H^+$ (not NADH). The remaining reactions known as **non-oxidative pathway**, convert Ribulose-5-phosphate molecules to various intermediates such as Ribose-5-phosphate(5C), Xylulose-5-phosphate(5C), Glyceraldehyde-3-phosphate(3C), Sedoheptulose-7-Phosphate (7C), and Erythrose-4-phosphate (4C). Finally, five molecules of glucose-6-phosphate is regene-rated (Figure 14.16). The overall reaction is:

**6 x Glucose-6-Phosphate + 12NADP$^+$ + 6$H_2O$**

↓

**5 x Glucose-6-Phosphate + 6$CO_2$ + Pi + 12NADPH +12$H^+$**

The net result of complete oxidation of one glucose-6-phosphate yield $6CO_2$ and 12NADPH + $H^+$. The oxidative pentose



XIth_Botany_Chapter_14_EM.indd 303 05-01-2023 18:30:27